

Profile

Mechanical Engineering student with hands-on experience in design and aerospace, skilled in CAD and project coordination, with a strong interest in sustainable engineering solutions.

Education

California State University - Fresno, Lyles College of Engineering
Bachelor of Science in Mechanical Engineering | Minor in Aerospace Engineering
Expected Graduation: Spring 2027 | **GPA:** 3.52

Technical Skills

Engineering & Analysis:

Systems Engineering | CFD | FEA | Numerical Methods |
Engineering Testing (MTS)

Software & Tools:

SolidWorks | PTC Creo | Siemens NX | ANSYS | MATLAB |
Simulink | Python | AutoCAD | Revit | SketchUp | ArtiosCAD

Hardware & Systems:

RF Systems | PCB Design | Arduino | Data Acquisition |
Instrumentation

Programming:

Python | MATLAB | C++

Manufacturing:

CNC Machining | Manual Machining

Work Experience & Research

Systems Engineering Intern, NASA Armstrong Flight Research Center, Edwards, CA

June 2026 – August 2026

- Developing a quantum sensor prototype integrating hardware and software components
- Contributing to RF, optics, and PCB-related system design and testing
- Supporting mechanical design and system integration using CAD and experimental validation
- Collaborating with a multidisciplinary team to iterate and refine system performance

Rotorcraft Aeromechanics Intern, NASA Ames Research Center, Moffett Field, CA

June 2025 – August 2025

- Developed a Python based rotor power calculator with a custom GUI using interpolated CAMRAD II power coefficient tables to evaluate rotorcraft performance.
- Automated airfoil slicing from 3D rotor blade geometry in Rhino using rhinoscriptsyntax to support aerodynamic analysis workflows.
- Supported research on next generation vertical flight vehicles by applying CFD, aeroelastic analysis, and wind tunnel testing to assess performance, noise, vibration, and ride quality.

Engineering Intern, Duncan Enterprises (iLoveToCreate), Fresno, CA

December 2023 – May 2025

- Created packaging specifications and engineering drawings for new product lines using SolidWorks and ArtiosCAD.
- Conducted time studies and data analysis to support production improvements and efficiency reporting.
- Wrote standardized packaging instructions and collaborated with Operations and Quality leadership.

Projects

Propulsion Team Member, NASA Student Launch Competition, California State University, Fresno

March 2026 – Present

- Contributing to the design of an N₂O/IPA liquid bipropellant rocket engine using the Mojave Sphinx framework.
- Performing propulsion calculations and supporting CAD modeling in SolidWorks.
- Assisting with manufacturing, assembly, and static fire test preparation.

Chassis and Propulsion Team Member, MATE ROV Competition, California State University, Fresno

November 2025 – Present

- Designed and evaluated a six thruster propulsion layout to enable full 6 DOF control and station keeping in flowing water.
- Contributed to chassis layout, thruster placement, and shroud considerations to improve stability, maneuverability, and safety.
- Developed tether routing and strain relief concepts to transfer loads into the chassis and protect onboard electronics.

Leadership and Activities

- Society of Women Engineers (SWE): Professional Development
- Society of Hispanic Professional Engineers (SHPE): Secretary